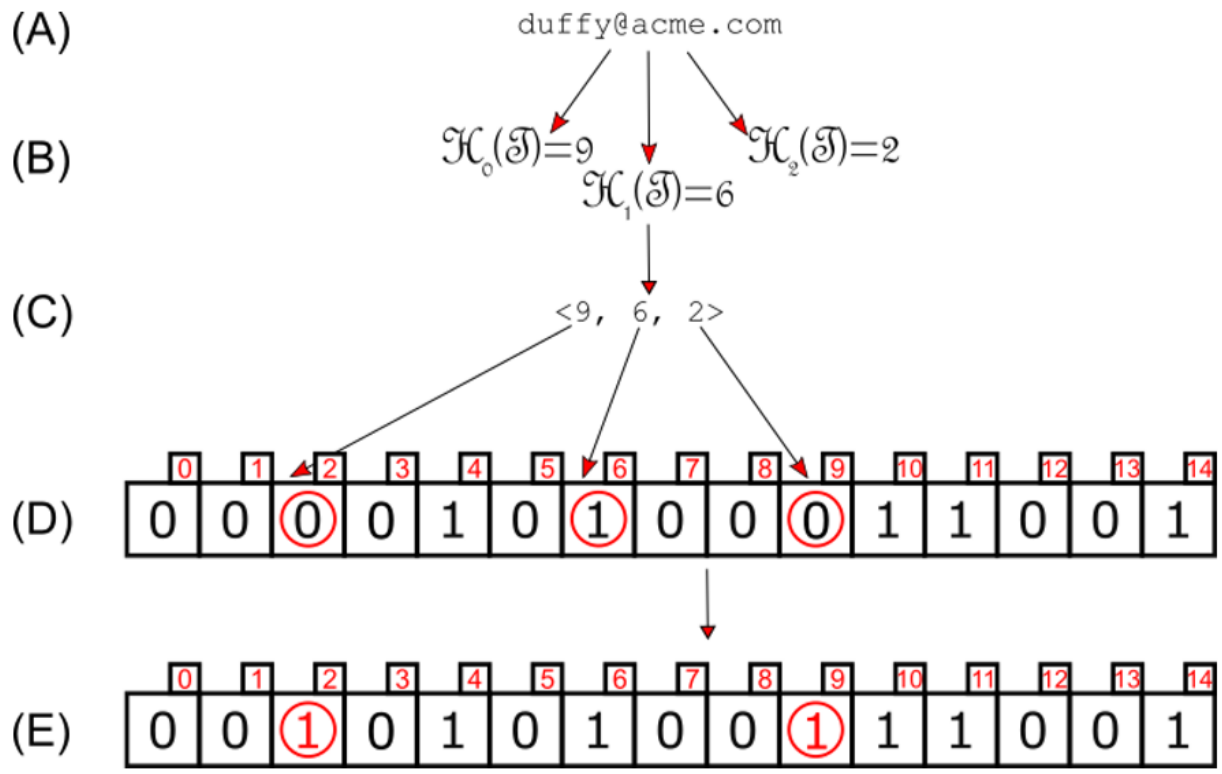


Figure 4.5 Adding a new entry to a Bloom filter, step by step.
(A) We start with the email we would like to check, "`duffy@acme.com`".
(B) The key (our email) is processed through our set of hash functions: in our example we assume `k=3`, so we'll have 3 different hash functions, H_0, H_1 and H_2.
(C) Each hash function produces an index for our binary array: in this case the three indices might be, for instance, `<9, 6, 2>`. Notice how these indices are the same as in figure 4.4: hash functions are deterministic, though their results might seem random.
(D) We access the elements of the binary array at those indices: the first element, at index `2`, is `0`. The other bits are `0` and `1` respectively.